Day 4 – Regression & Comparing Models

So far we've predicted **categories**.

Now introduce another major type of ML:

Regression

Regression predicts a numerical value.

For example:

Temperature
Voltage
Current
RPM
↓
Predicted Motor Temperature

Or:

Motor Age
Operating Hours
Temperature
Vibration
↓
Predicted Remaining Operating Hours

Learn

Study:

- Classification vs Regression
- Linear Regression
- Decision Tree Regression
- Prediction of continuous values

Practical Work

Use a simple dataset such as the California Housing dataset from scikit-learn.

Build a model that predicts house prices.

Don't spend too much time worrying about the dataset itself. The purpose is to understand **regression**.

Program 1 – Linear Regression

Steps:

1. Load dataset.
2. Separate features and target.
3. Split into training/testing data.
4. Create Linear Regression model.
5. Train it.
6. Make predictions.
7. Evaluate the predictions.

Program 2 – Compare

Try another model:

Decision Tree Regressor

Compare the two models.

Write:

Which model performed better?

Why do you think it performed better?

Git Task

Commit and push today's work.

Day 5 – Mini Machine Learning Project

Now bring everything together.

Student Performance Prediction

Build a small ML project that predicts whether a student will **Pass or Fail**.

You can create a small CSV dataset yourself.

Example:

```
StudyHours,Attendance,Assignments,Result

2,60,50,Fail
3,65,55,Fail
4,70,65,Fail
5,75,70,Pass
6,80,75,Pass
7,85,80,Pass
8,90,90,Pass
```

You should have **more records than the example above**.

Requirements

1. Load the dataset

Use Pandas.

---

2. Explore the dataset

Display:

- Number of records
- Columns
- First few records
- Missing values

---

3. Prepare the data

Identify:

Features

StudyHours
Attendance
Assignments

Target

Result

---

4. Train a Model

Use:

Decision Tree Classifier

Split the dataset into:

Training Data
Testing Data

Train the model.

---

5. Make Predictions

Give the model a new student's information.

For example:

Study Hours: 6
Attendance: 82
Assignments: 75

Ask the model:

PASS or FAIL?

---

6. Evaluate

Calculate:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

---

7. Explain Your Model

Create:

README.md

Explain:

- What problem you were solving.
- What your features were.
- What your target was.
- What model you used.
- How you trained it.
- How accurate it was.
- What you learned.
- What could be improved.

Questions

Basic

What is Machine Learning?

Understanding

What is a feature?

What is a label?

Practical

Why did we split our dataset into training and testing data?

Model

What does `fit()` do?

What does `predict()` do?

Evaluation

Your model has 95% accuracy. Does that mean it is perfect?